MAT 167 Final Project: Google's PageRank Algorithm

UC Davis · Winter Quarter 2026 · Instructor: DP Varn
Network: Google Network #10 · Keywords: Trees Set #3

How This Notebook Is Organized

Each section below follows the numbered steps from the project instructions exactly:

Section	Content
1	Imports & Setup
2	Define the Network & Build Raw Google Matrix G
3	Eigendecomposition of Raw G (Steps 2–3)
4	Fix Dangling Nodes (Step 4)
5	Eigendecomposition of Fixed G (Step 5)
6	Normalize π and Rank Pages (Steps 6–7)
7	Power Method (Step 8)
8	Teleportation Matrix $\tilde{G}$ (Step 9)
9	Eigendecomposition of $\tilde{G}$ (Step 10)
10	Compare Rankings: G vs $\tilde{G}$ (Step 11)
11	Term-Document Matrix T (Step 12)
12	Search Queries (Steps 13–15)
13	Summary of All Results

Section 1 — Imports & Setup

We need three libraries:

- `numpy`: all matrix and vector operations (building G, normalizing π, argsort for ranking)
- `scipy.linalg`: `eig()` for full eigenvalue decomposition of a square matrix
- `matplotlib`: bar charts, convergence plots, heatmaps for the presentation

No other libraries are needed for the math. Everything in PageRank reduces to matrix multiplication and eigendecomposition.

In [1]:
```
import numpy as np
from scipy.linalg import eig
import matplotlib.pyplot as plt
import matplotlib.colors as mcolors

# Make plots look clean and readable
plt.rcParams.update({
    'figure.figsize': (10, 5),
    'font.size': 12,
    'axes.titlesize': 14,
    'axes.labelsize': 12,
})

print("All imports successful.")
print(f"NumPy version: {np.__version__}")
```

```
All imports successful.
NumPy version: 1.26.4
```

Section 2 — Define the Network & Build Raw Google Matrix G

(Project Instruction Step 1)

What the Google Matrix G is:
G is an `n × n` matrix where `n = 12` (one row and column per page).
The entry `G[i][j]` gives the probability that a random surfer on page `i` clicks a link to page `j`:

$$G_{ij} = \begin{cases} \frac{1}{|P_i|} & \text{if page } i \text{ links to page } j \\ 0 & \text{otherwise} \end{cases}$$

where $|P_i|$ is the number of outlinks from page i.
Each outlink gets equal weight: page i's "vote" is divided equally among all pages it links to.

Lecture reference (Saito Lec 25, Slide 6):
The Google matrix $G = (g_{ij})$ with $g_{ij} = 1/|P_i|$ if P_i links to P_j, else 0.

Node index convention (fixed throughout):

A	B	C	D	E	F	G	H	I	J	K	L
0	1	2	3	4	5	6	7	8	9	10	11

Validation rule: every row of G must sum to either **1.0** (page has outlinks) or **0.0** (dangling node with no outlinks). Any other value means a bug in the adjacency list.

In [3]:
```
# Node setup
n = 12
labels = list('ABCDEFGHIJKL')   # labels[0]='A', labels[1]='B', ...

# Page titles (for readable output)
page_titles = {
    'A': 'Trees For All',
    'B': 'The Weeping Willow',
    'C': 'Leaf Fall',
    'D': 'Shady Place',
    'E': 'The Prickly Pine',
    'F': 'Good Wood',
    'G': 'Lumber Jack Man',
    'H': 'Timber!',
    'I': 'Fine Grains',
    'J': 'The Warped Board',
    'K': 'Evergreen Plants',
    'L': 'Sappy Planks',
}

# Adjacency list
# adj[i] = list of node indices that page i links TO (outlinks)
# Read directly from GoogleNet10 diagram.
# VERIFY THIS AGAINST THE HAND-DRAWN DIAGRAM BEFORE RUNNING.
adj = {
    0:  [1, 5],          # A → B, F
    1:  [0, 2, 5, 6],    # B → A, C, F, G
    2:  [3, 6],          # C → D, G
    3:  [0, 7],          # D → A, H
    4:  [0, 5, 8],       # E → A, F, I
    5:  [0, 8, 9],       # F → A, I, J
    6:  [3, 5, 7, 10],   # G → D, F, H, K
    7:  [11],            # H → L
    8:  [4, 5, 9],       # I → E, F, J
    9:  [6],             # J → G
    10: [9],             # K → J
    11: [3, 10],         # L → D, K
}

# Build raw Google matrix
def build_google_matrix(adj, n):
    """
    Build the raw Google matrix G from an adjacency list.
    G[i][j] = 1/len(outlinks_from_i)  if i links to j
    G[i][j] = 0                        otherwise
    Rows with no outlinks stay all-zero (dangling nodes).
    """
    G = np.zeros((n, n))
    for i in range(n):
        outlinks = adj.get(i, [])
        if len(outlinks) > 0:
            for j in outlinks:
```

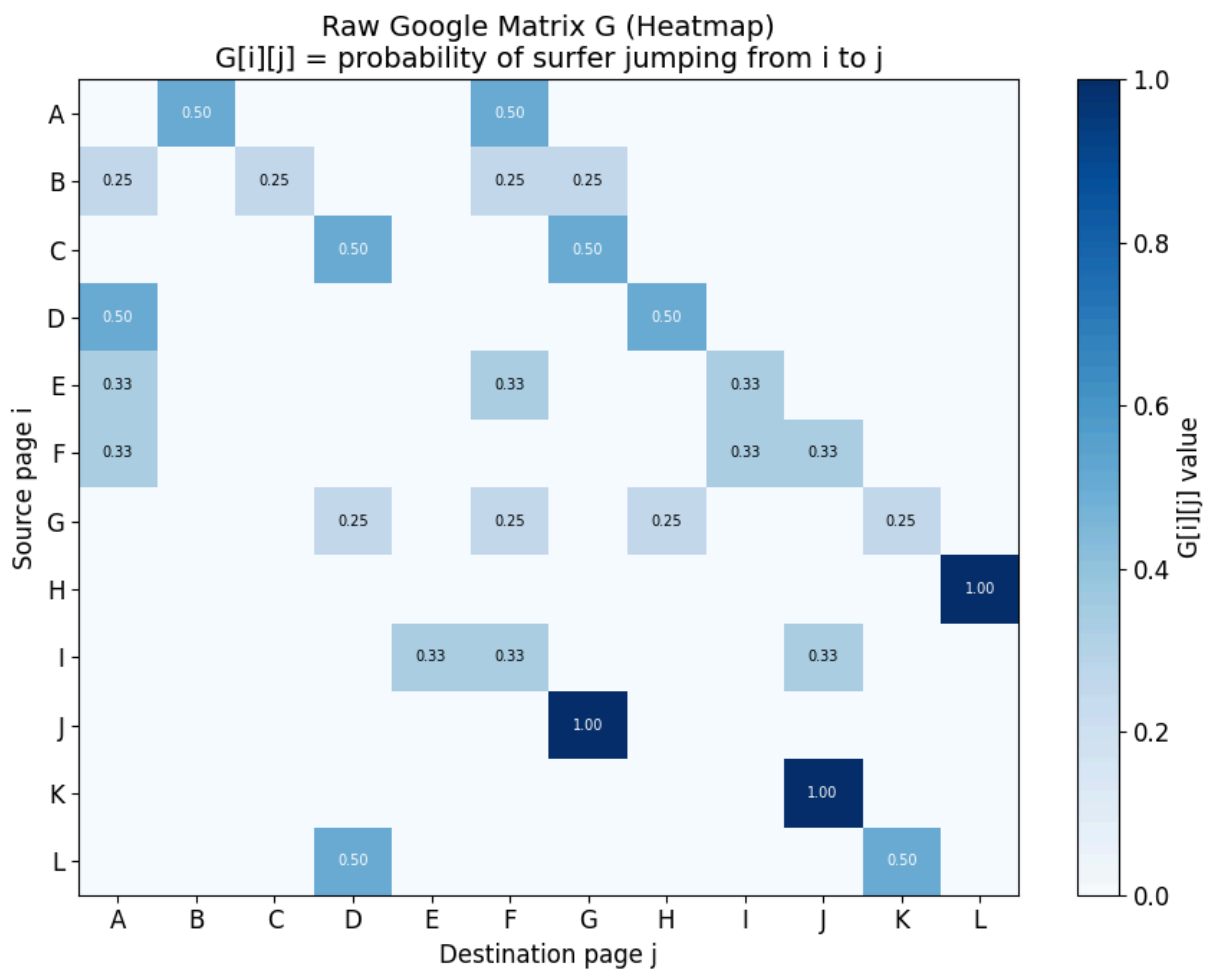
```
                G[i][j] = 1.0 / len(outlinks)
    return G

G_raw = build_google_matrix(adj, n)

# Validation
row_sums = G_raw.sum(axis=1)
print("Row sums of G_raw (must be 0.0 or 1.0):")
for i, s in enumerate(row_sums):
    status = "✓" if abs(s - 1.0) < 1e-10 or abs(s) < 1e-10 else "✗ ERROR"
    print(f"  Page {labels[i]}: {s:.6f}  {status}")

dangling = [labels[i] for i, s in enumerate(row_sums) if abs(s) < 1e-10]
print(f"\nDangling nodes (zero row sum): {dangling if dangling else 'None'}"
```

```
Row sums of G_raw (must be 0.0 or 1.0):
  Page A: 1.000000  ✓
  Page B: 1.000000  ✓
  Page C: 1.000000  ✓
  Page D: 1.000000  ✓
  Page E: 1.000000  ✓
  Page F: 1.000000  ✓
  Page G: 1.000000  ✓
  Page H: 1.000000  ✓
  Page I: 1.000000  ✓
  Page J: 1.000000  ✓
  Page K: 1.000000  ✓
  Page L: 1.000000  ✓

Dangling nodes (zero row sum): None
```

In [4]:

```
# Visualize G_raw as a heatmap
# Slide 13

fig, ax = plt.subplots(figsize=(9, 7))
im = ax.imshow(G_raw, cmap='Blues', aspect='auto', vmin=0)
plt.colorbar(im, ax=ax, label='G[i][j] value')

ax.set_xticks(range(n))
ax.set_yticks(range(n))
ax.set_xticklabels(labels)
ax.set_yticklabels(labels)
ax.set_xlabel('Destination page j')
ax.set_ylabel('Source page i')
ax.set_title('Raw Google Matrix G (Heatmap)\nG[i][j] = probability of surfer

# Annotate non-zero cells with their values
for i in range(n):
    for j in range(n):
        if G_raw[i, j] > 0:
            ax.text(j, i, f'{G_raw[i,j]:.2f}', ha='center', va='center',
                    fontsize=7, color='white' if G_raw[i,j] > 0.4 else 'blac

plt.tight_layout()
plt.savefig('G_raw_heatmap.png', dpi=150, bbox_inches='tight')
```

```
plt.show()
print("Saved: G_raw_heatmap.png")
```




```
Saved: G_raw_heatmap.png
```

Section 3 — Eigendecomposition of Raw G

(Project Instruction Steps 2–3)

The eigenvalue equation we are solving:

$$\pi^T G = \pi^T \qquad \Leftrightarrow \qquad G^T \pi = \pi$$

We want the vector π such that multiplying by G leaves it unchanged. This is the **eigenvector for eigenvalue** $\lambda = 1$.
In `scipy`, `eig(G.T)` returns the **right eigenvectors** of G^T, which are the same as the **left eigenvectors** of G — exactly what we need.

Expected result with raw G:
If there are dangling nodes, G is not fully row-stochastic, so $\lambda_{\max} < 1$.
Saito Lecture 25, Slide 7 shows the 6-node example where $\lambda_{\max} = 0.9207$ with the raw G.

This is a *problem* — a dominant eigenvalue $\neq 1$ means the power iteration does not converge to a valid stationary distribution. **We need to fix G first.**

In [5]:
```
def compute_pagerank_eig(G, label='G'):
    """
    Compute PageRank via direct eigendecomposition of G^T.
    Solves: G^T @ pi = lambda * pi
    Returns: (pi, dominant_eigenvalue, all_eigenvalues)

    Steps:
    1. Compute all eigenvalues and eigenvectors of G^T
    2. Find the index of the largest real eigenvalue
    3. Extract that eigenvector, take absolute value, L1-normalize
    """
    # eig(G.T) returns:
    #   eigenvalues: array of n complex numbers
    #   eigenvectors: n×n matrix, COLUMNS are eigenvectors
    eigenvalues, eigenvectors = eig(G.T)

    # Find the index of the eigenvalue with the largest real part
    # For a proper stochastic matrix, this will be 1.0 (or very close)
    idx = np.argmax(np.real(eigenvalues))
    lambda_max = np.real(eigenvalues[idx])

    # Extract the corresponding eigenvector (column idx of the matrix)
    pi = np.real(eigenvectors[:, idx])

    # Eigenvectors can have arbitrary sign — take absolute value
    # to ensure all entries are non-negative (required for a probability vec
    pi = np.abs(pi)

    # L1-normalize: divide by sum so that sum(pi) = 1
    # This makes pi a proper probability distribution
    pi = pi / pi.sum()

    print(f"\n{'='*50}")
    print(f"Eigendecomposition of {label}")
    print(f"{'='*50}")
    print(f"  Dominant eigenvalue (λ_max): {lambda_max:.8f}")
    print(f"  All eigenvalues (real parts): {sorted(np.real(eigenvalues), re
    print(f"  sum(π) after normalization:  {pi.sum():.8f}  (must be 1.0)")
    print(f"  min(π):                      {pi.min():.8f}  (must be ≥ 0)")

    return pi, lambda_max, eigenvalues


# Step 2: Eigendecomposition of raw G
pi_raw, lambda_raw, evals_raw = compute_pagerank_eig(G_raw, 'G_raw')

print("\nPageRank vector from raw G (before any fixes):")
print(f"  {'Page':<6} {'Title':<22} {'π value':<12}")
print(f"  {'-'*42}")
for i in range(n):
    print(f"  {labels[i]:<6} {page_titles[labels[i]]:<22} {pi_raw[i]:.6f}")
```

```
print(f"\n λ_max = {lambda_raw:.4f} (expected 1.0 — this will be fixed in Se
```

```
==================================================
Eigendecomposition of G_raw
==================================================
  Dominant eigenvalue (λ_max): 1.00000000
  All eigenvalues (real parts): [1.0000000000000018, 0.625273785016588, 0.39
66543094646142, 0.0470618727101512, 0.0470618727101512, -0.0538875835029308
8, -0.05388758350293088, -0.21774244358192865, -0.21774244358192865, -0.4794
9354397940336, -0.47949354397940336, -0.6138046977729764]
  sum(π) after normalization:  1.00000000  (must be 1.0)
  min(π):                      0.01312019  (must be ≥ 0)

PageRank vector from raw G (before any fixes):
  Page   Title                  π value
  ----------------------------------------
  A      Trees For All          0.104962
  B      The Weeping Willow     0.052481
  C      Leaf Fall              0.013120
  D      Shady Place            0.088674
  E      The Prickly Pine       0.015835
  F      Good Wood              0.126678
  G      Lumber Jack Man        0.159855
  H      Timber!                0.084301
  I      Fine Grains            0.047504
  J      The Warped Board       0.140175
  K      Evergreen Plants       0.082114
  L      Sappy Planks           0.084301

 λ_max = 1.0000 (expected 1.0 — this will be fixed in Section 4)
```

Section 4 — Fix Dangling Nodes → Make G Row-Stochastic

(Project Instruction Step 4)

The problem: A dangling node (no outlinks) gives an all-zero row in G.
Row sums of 0 make G *not* row-stochastic, which causes $\lambda_{\max} < 1$.

The fix (Saito Lec 25, Slide 9):
For every dangling node i, replace its row with $[1/n, 1/n, \ldots, 1/n]$.
This models the random surfer teleporting to any page uniformly when they reach a dead end.

After this fix, **every row must sum to exactly 1.0** — G is now fully row-stochastic.

In [6]:
```
def fix_dangling_nodes(G, n, labels):
    """
    Fix dangling nodes by replacing all-zero rows with uniform row [1/n, ...
```

```
    Returns a new matrix (does not mutate the input).
    After fix: every row sums to exactly 1.0 → G is row-stochastic.
    """
    G_fixed = G.copy()   # work on a copy so G_raw is preserved
    fixed_nodes = []

    for i in range(n):
        if G_fixed[i].sum() < 1e-10:   # row is all zeros → dangling
            G_fixed[i] = 1.0 / n       # replace with uniform
            fixed_nodes.append(labels[i])

    if fixed_nodes:
        print(f"Fixed dangling nodes: {fixed_nodes}")
    else:
        print("No dangling nodes found — G_raw is already fully row-stochast

    return G_fixed


G_fixed = fix_dangling_nodes(G_raw, n, labels)

# Verify all row sums are now 1.0
row_sums_fixed = G_fixed.sum(axis=1)
print("\nRow sums of G_fixed (all must be 1.0):")
all_ok = True
for i, s in enumerate(row_sums_fixed):
    ok = abs(s - 1.0) < 1e-10
    if not ok:
        all_ok = False
    print(f"  Page {labels[i]}: {s:.8f}  {'✓' if ok else '✗ ERROR'}")
print(f"\nAll rows sum to 1? {'✓ Yes' if all_ok else '✗ No — check adj dicti
```

```
No dangling nodes found — G_raw is already fully row-stochastic.

Row sums of G_fixed (all must be 1.0):
  Page A: 1.00000000  ✓
  Page B: 1.00000000  ✓
  Page C: 1.00000000  ✓
  Page D: 1.00000000  ✓
  Page E: 1.00000000  ✓
  Page F: 1.00000000  ✓
  Page G: 1.00000000  ✓
  Page H: 1.00000000  ✓
  Page I: 1.00000000  ✓
  Page J: 1.00000000  ✓
  Page K: 1.00000000  ✓
  Page L: 1.00000000  ✓

All rows sum to 1? ✓ Yes
```

Section 5 — Eigendecomposition of Fixed G

(Project Instruction Step 5)

Expected result: Now that G is row-stochastic, the **Perron-Frobenius Theorem** guarantees:

1. $\lambda_{\max} = 1$ exactly
2. The corresponding eigenvector has **all non-negative entries**
3. The eigenvector is unique (up to scaling) if the Markov chain is irreducible

Core mathematical result that makes PageRank work.
(Saito Lec 25, Slide 10: after fixing the dangling node, λ_max = 1 and the eigenvector is non-negative.)

In [7]:
```
# Eigendecomposition of G_fixed
pi_fixed, lambda_fixed, evals_fixed = compute_pagerank_eig(G_fixed, 'G_fixed

print(f"\n✓ λ_max = {lambda_fixed:.8f}  (should be 1.0)")
print(f"✓ All entries non-negative: {(pi_fixed >= -1e-12).all()}")
```

```
==================================================
Eigendecomposition of G_fixed
==================================================
  Dominant eigenvalue (λ_max): 1.00000000
  All eigenvalues (real parts): [1.0000000000000018, 0.625273785016588, 0.39
66543094646142, 0.0470618727101512, 0.0470618727101512, -0.0538875835029308
8, -0.05388758350293088, -0.21774244358192865, -0.21774244358192865, -0.4794
9354397940336, -0.47949354397940336, -0.6138046977729764]
  sum(π) after normalization:  1.00000000  (must be 1.0)
  min(π):                      0.01312019  (must be ≥ 0)

✓ λ_max = 1.00000000  (should be 1.0)
✓ All entries non-negative: True
```

Section 6 — Normalize π and Rank Pages

(Project Instruction Steps 6–7)

Step 6 — Normalize:
The eigenvector is already normalized inside `compute_pagerank_eig()` (we divide by `pi.sum()`).
This ensures $\|\pi\|_1 = \sum_i \pi_i = 1$, so π is a valid probability distribution.

Step 7 — Rank:
Sort the entries of π in **non-increasing order** (largest to smallest).
The page with the highest π_i is the most important page in the network.

In [8]:
```
def rank_pages(pi, labels, page_titles, header='PageRank Ranking'):
    """
    Sort pages by PageRank in descending order and print a formatted table.
    Returns the sorted index array.
```

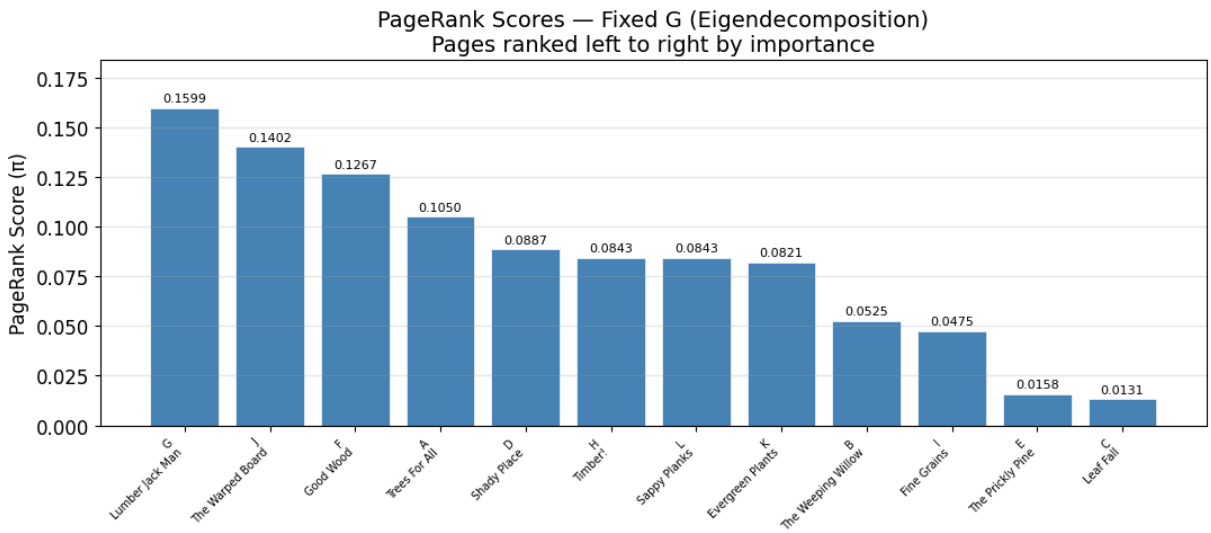
```
    """
    sorted_idx = np.argsort(-pi)   # argsort of -pi gives descending order

    print(f"\n{header}")
    print(f"{'Rank':<6} {'Page':<6} {'Title':<25} {'π value':<12} {'% of tot
    print("-" * 65)
    for rank, i in enumerate(sorted_idx, start=1):
        pct = pi[i] * 100
        bar = '▇' * int(pct * 2)   # simple ASCII bar
        print(f"  #{rank:<4} {labels[i]:<6} {page_titles[labels[i]]:<25} {pi

    print(f"\n  sum(π) = {pi.sum():.8f}  (must be 1.0)")
    return sorted_idx


print("Step 6: π is already L1-normalized (done inside compute_pagerank_eig)
print(f"  Verification: sum(pi_fixed) = {pi_fixed.sum():.10f}")

print("\nStep 7: Ranking pages by PageRank score")
sorted_fixed = rank_pages(pi_fixed, labels, page_titles, 'PageRank from G_fi
```

```
Step 6: π is already L1-normalized (done inside compute_pagerank_eig).
  Verification: sum(pi_fixed) = 1.0000000000

Step 7: Ranking pages by PageRank score

PageRank from G_fixed (Eigendecomposition)
Rank   Page   Title                     π value      % of total
-----------------------------------------------------------------
  #1    G      Lumber Jack Man           0.159855     15.99%  ▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇▇
  #2    J      The Warped Board          0.140175     14.02%  ▇▇▇▇▇▇▇▇▇▇▇▇▇▇
  #3    F      Good Wood                 0.126678     12.67%  ▇▇▇▇▇▇▇▇▇▇▇▇
  #4    A      Trees For All             0.104962     10.50%  ▇▇▇▇▇▇▇▇▇▇▇▇▇
  #5    D      Shady Place               0.088674     8.87%  ▇▇▇▇▇▇▇▇▇▇▇▇▇▇
  #6    H      Timber!                   0.084301     8.43%  ▇▇▇▇▇▇▇▇▇▇▇▇▇▇
  #7    L      Sappy Planks              0.084301     8.43%  ▇▇▇▇▇▇▇▇▇▇▇▇▇▇
  #8    K      Evergreen Plants          0.082114     8.21%  ▇▇▇▇▇▇▇▇▇▇▇▇▇▇
  #9    B      The Weeping Willow        0.052481     5.25%  ▇▇▇▇▇▇▇▇▇▇
  #10   I      Fine Grains               0.047504     4.75%  ▇▇▇▇▇▇▇▇▇
  #11   E      The Prickly Pine          0.015835     1.58%  ▇▇▇
  #12   C      Leaf Fall                 0.013120     1.31%  ▇▇

  sum(π) = 1.00000000  (must be 1.0)
```

In [9]:

```
# Bar chart of PageRank from G_fixed
# Slide 14

fig, ax = plt.subplots(figsize=(11, 5))
```

```
sorted_idx = np.argsort(-pi_fixed)
page_labels_sorted = [f"{labels[i]}\n{page_titles[labels[i]]}" for i in sort
pi_sorted = pi_fixed[sorted_idx]

bars = ax.bar(range(n), pi_sorted, color='steelblue', edgecolor='white', lir

# Annotate each bar with the exact value
for bar, val in zip(bars, pi_sorted):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.002,
            f'{val:.4f}', ha='center', va='bottom', fontsize=8)

ax.set_xticks(range(n))
ax.set_xticklabels(page_labels_sorted, fontsize=7, rotation=45, ha='right')
ax.set_ylabel('PageRank Score (π)')
ax.set_title('PageRank Scores — Fixed G (Eigendecomposition)\nPages ranked l
ax.set_ylim(0, max(pi_sorted) * 1.15)
ax.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.savefig('pagerank_fixed_G.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: pagerank_fixed_G.png")
```




```
Saved: pagerank_fixed_G.png
```

Section 7 — Power Method

(Project Instruction Step 8)

Why the Power Method?

Direct eigendecomposition (`scipy.linalg.eig`) requires $O(n^3)$ operations, fine for $n = 12$, but completely infeasible for Google's $n = 3 \times 10^8$ pages.

The Power Method is iterative: it only ever computes matrix-vector products, which are cheap and can exploit sparsity.

The update rule (Saito Lec 25, Slide 6):

$$\pi_j^T = \pi_{j-1}^T \cdot G, \qquad \text{then normalize } \pi_j = \pi_j / \|\pi_j\|_1$$

Starting point: $\pi_0 = \frac{1}{n}\mathbf{1}_n$ (uniform — every page equally likely).

Stopping rule: Stop when $\|\pi_j - \pi_{j-1}\|_1 \leq \varepsilon$ for some tolerance ε.

Key note about notation:
`pi @ G` (row vector times G on the right) implements $\pi^T G$.
Do **not** write `G @ pi` — that would compute $G\pi$ which is the update for a right eigenvector, not what we want.

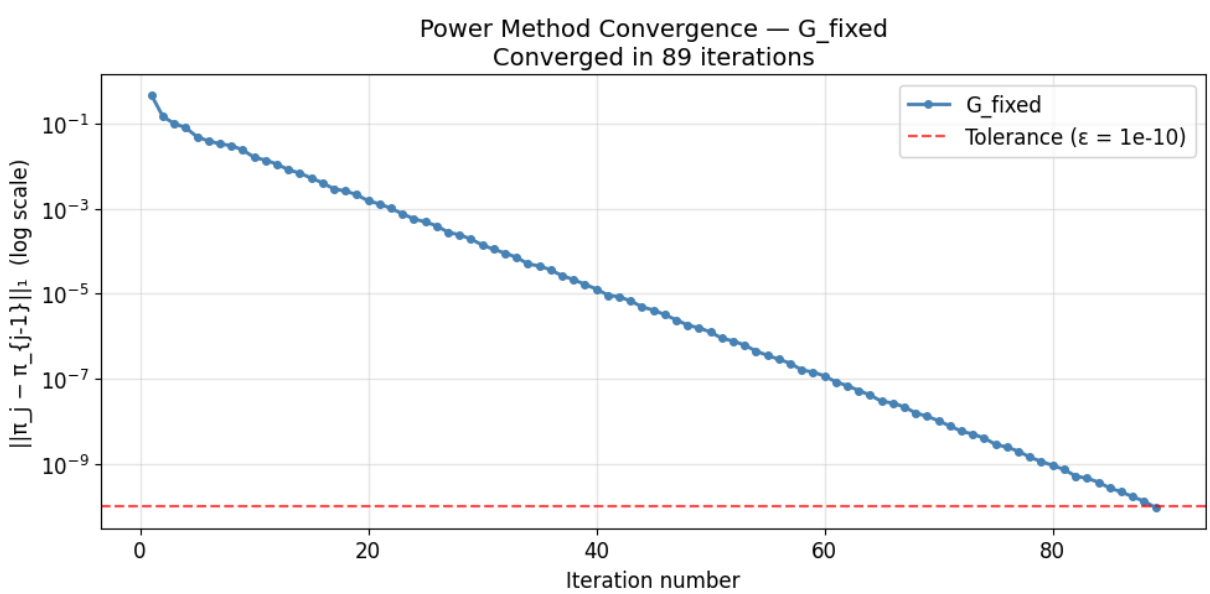
In [10]:
```
def power_method(G, tol=1e-10, max_iter=1000, label='G'):
    """
    Find the dominant left eigenvector of G using power iteration.

    Update rule:  π_j^T = π_{j-1}^T @ G   (row vector times G)
    Stop when:    ||π_j - π_{j-1}||_1 <= tol

    Returns: (pi, iteration_count, convergence_history)
    """
    n = G.shape[0]

    # π₀ = [1/n, 1/n, ..., 1/n] — uniform start, already L1-normalized
    pi = np.ones(n) / n
    history = []          # stores ||π_j - π_{j-1}||_1 at each step

    for k in range(1, max_iter + 1):
        # Core update: π_j^T = π_{j-1}^T @ G
        # 'pi' is a 1D numpy array treated as a row vector
        pi_new = pi @ G

        # Re-normalize to prevent floating-point drift
        pi_new = pi_new / pi_new.sum()

        # Compute L1 norm of the change (convergence measure)
        delta = np.linalg.norm(pi_new - pi, ord=1)
        history.append(delta)

        if delta <= tol:
            print(f"Power Method [{label}]: converged in {k} iterations  (fi
            return pi_new, k, history

        pi = pi_new

    print(f"Power Method [{label}]: WARNING — did not converge in {max_iter}
    return pi, max_iter, history


# Run on G_fixed
pi_power_fixed, iters_fixed, hist_fixed = power_method(G_fixed, label='G_fix
```

```
# Compare power method result with eigendecomp result
print(f"\nMax difference between power method and eigendecomp results:")
print(f"  ||π_power − π_eig||_∞ = {np.max(np.abs(pi_power_fixed − pi_fixed))
print("  (Should be < 1e−8 — tiny numerical difference only)")
```

```
Power Method [G_fixed]: converged in 89 iterations  (final δ = 9.58e−11)

Max difference between power method and eigendecomp results:
  ||π_power − π_eig||_∞ = 1.37e−11
  (Should be < 1e−8 — tiny numerical difference only)
```

In [11]:

```
# Convergence plot
# Slide 15
# Log-scale shows the exponential convergence rate clearly.

fig, ax = plt.subplots(figsize=(10, 5))

ax.semilogy(range(1, len(hist_fixed) + 1), hist_fixed,
            color='steelblue', linewidth=2, marker='o', markersize=4,
            label='G_fixed')

ax.axhline(y=1e−10, color='red', linestyle='−−', alpha=0.7, label='Tolerance

ax.set_xlabel('Iteration number')
ax.set_ylabel('||π_j − π_{j−1}||₁  (log scale)')
ax.set_title(f'Power Method Convergence — G_fixed\n'
             f'Converged in {iters_fixed} iterations')
ax.legend()
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('convergence_G_fixed.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: convergence_G_fixed.png")
```




```
Saved: convergence_G_fixed.png
```

Section 8 — Build the Teleportation Matrix $\tilde{G}$

(Project Instruction Step 9)

Why $\tilde{G}$ is needed (Saito Lec 25, Slides 14–15):
Even after fixing dangling nodes, the graph may not be *strongly connected* (there may be subsets of pages that trap the random surfer permanently). This is called a **reducible Markov chain**, and it produces non-unique or zero-valued PageRank entries for isolated pages.

The fix — teleportation:
Brin and Page introduced a "damping factor" $\alpha = 0.85$:

$$\tilde{G} = \alpha G + (1-\alpha)E, \qquad E = \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^T$$

- With probability $\alpha = 0.85$: the surfer follows a link normally
- With probability $1-\alpha = 0.15$: the surfer teleports to any random page (like typing a URL)

$\tilde{G}$ has **all positive entries** (since every entry of $E = 1/n > 0$), which makes it **irreducible**.
By Perron-Frobenius, an irreducible stochastic matrix has a **unique, fully positive** stationary distribution — the PageRank vector is now guaranteed to exist and be unique.

In [12]:
```
# Build G_tilde = alpha * G_fixed + (1 - alpha) * E
alpha = 0.85
E = np.ones((n, n)) / n         # E[i][j] = 1/n for all i,j
G_tilde = alpha * G_fixed + (1 - alpha) * E

# Verification
print(f"alpha = {alpha},  (1 - alpha) = {1 - alpha}")
print(f"\nE matrix: every entry = {1/n:.6f} (= 1/{n})")

row_sums_tilde = G_tilde.sum(axis=1)
print(f"\nRow sums of G_tilde (all must be 1.0):")
all_ok = all(abs(s - 1.0) < 1e-10 for s in row_sums_tilde)
print(f"  Min row sum: {row_sums_tilde.min():.10f}")
print(f"  Max row sum: {row_sums_tilde.max():.10f}")
print(f"  All rows sum to 1? {'✓ Yes' if all_ok else '✗ No'}")

print(f"\nMin entry of G_tilde: {G_tilde.min():.6f}  (must be > 0 for irredu
print(f"  → G_tilde is {'✓ irreducible' if G_tilde.min() > 0 else '✗ NOT irr
print(f"  → Perron-Frobenius guarantees unique positive PageRank vector.")
```

```
alpha = 0.85,  (1 - alpha) = 0.15000000000000002

E matrix: every entry = 0.083333 (= 1/12)

Row sums of G_tilde (all must be 1.0):
  Min row sum: 1.0000000000
  Max row sum: 1.0000000000
  All rows sum to 1? ✓ Yes

Min entry of G_tilde: 0.012500  (must be > 0 for irreducibility)
  → G_tilde is ✓ irreducible
  → Perron-Frobenius guarantees unique positive PageRank vector.
```

Section 9 — Eigendecomposition of $\tilde{G}$ (the True Google Matrix)

(Project Instruction Step 10)

This is the **final, authoritative PageRank**. We use both methods (eigendecomposition and power method) and verify they agree. The ranking from $\tilde{G}$ is used for all search queries.

In [14]:
```
# Eigendecomposition of G_tilde
pi_tilde, lambda_tilde, evals_tilde = compute_pagerank_eig(G_tilde, 'G_tilde

print(f"\n✓ λ_max = {lambda_tilde:.8f}  (should be 1.0)")
print(f"✓ All entries > 0: {(pi_tilde > 0).all()}  (guaranteed by irreducibi

sorted_tilde = rank_pages(pi_tilde, labels, page_titles, 'PageRank from G_ti
```

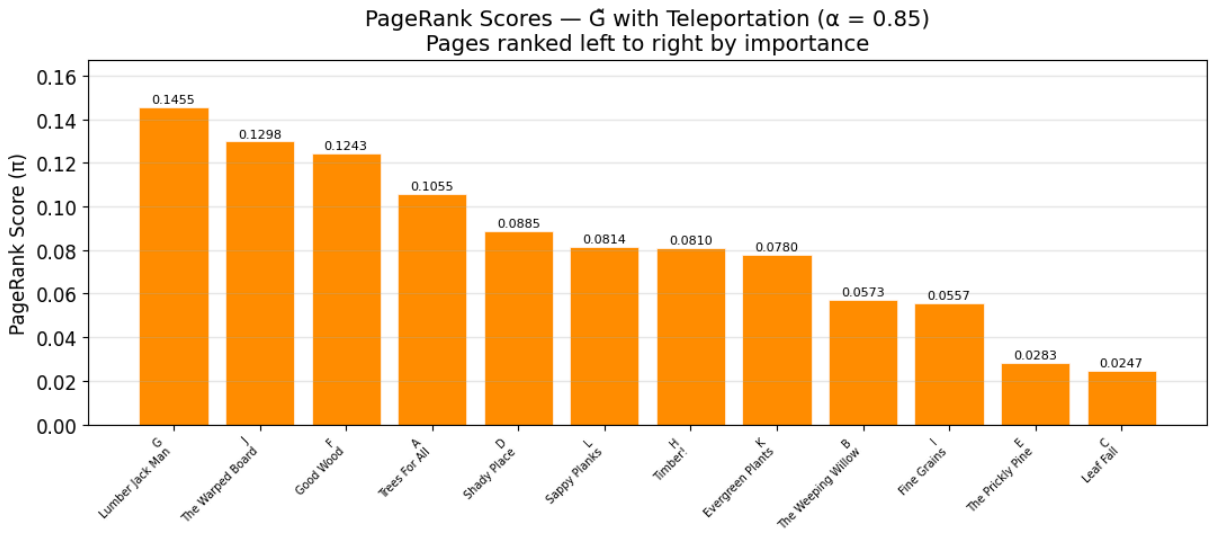
```
==================================================
Eigendecomposition of G_tilde
==================================================
  Dominant eigenvalue (λ_max): 1.00000000
  All eigenvalues (real parts): [1.0000000000000007, 0.5314827172640986, 0.3
371561630449218, 0.04000259180362876, 0.04000259180362876, -0.04580444597749
145, -0.04580444597749145, -0.1850810770446392, -0.1850810770446392, -0.4075
695123824933, -0.4075695123824933, -0.5217339931070326]
  sum(π) after normalization:  1.00000000  (must be 1.0)
  min(π):                      0.02468556  (must be ≥ 0)

✓ λ_max = 1.00000000  (should be 1.0)
✓ All entries > 0: True  (guaranteed by irreducibility)

PageRank from G_tilde
Rank   Page   Title                      π value      % of total
-------------------------------------------------------------------
  #1    G      Lumber Jack Man           0.145502     14.55%  ██████████
└──────────────┘
  #2    J      The Warped Board          0.129794     12.98%  ██████████
└──────────┘
  #3    F      Good Wood                 0.124250     12.43%  ██████████
└─────────┘
  #4    A      Trees For All             0.105515     10.55%  ██████████
└──────┘
  #5    D      Shady Place               0.088495     8.85%  ██████████
└┘
  #6    L      Sappy Planks              0.081375     8.14%  ██████████
└
  #7    H      Timber!                   0.081030     8.10%  ██████████
└
  #8    K      Evergreen Plants          0.078004     7.80%  ██████████
  #9    B      The Weeping Willow        0.057344     5.73%  ██████
  #10   I      Fine Grains               0.055719     5.57%  ██████
  #11   E      The Prickly Pine          0.028287     2.83%  ▌
  #12   C      Leaf Fall                 0.024686     2.47%  ▌

  sum(π) = 1.00000000  (must be 1.0)
```

In [15]:
```
# Power method on G_tilde (verify agreement)
pi_power_tilde, iters_tilde, hist_tilde = power_method(G_tilde, label='G_til

print(f"\nMax difference (power method vs eigendecomp) for G_tilde:")
print(f"  ||π_power - π_eig||_∞ = {np.max(np.abs(pi_power_tilde - pi_tilde))
```

```
Power Method [G_tilde]: converged in 54 iterations  (final δ = 6.95e-11)

Max difference (power method vs eigendecomp) for G_tilde:
  ||π_power - π_eig||_∞ = 5.98e-12
```

In [16]:
```
# Bar chart of PageRank from G_tilde
# Slide 16

fig, ax = plt.subplots(figsize=(11, 5))

sorted_idx_t = np.argsort(-pi_tilde)
```

```
page_labels_sorted_t = [f"{labels[i]}\n{page_titles[labels[i]]}" for i in sc
pi_tilde_sorted = pi_tilde[sorted_idx_t]

bars = ax.bar(range(n), pi_tilde_sorted, color='darkorange', edgecolor='whit
for bar, val in zip(bars, pi_tilde_sorted):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.001,
            f'{val:.4f}', ha='center', va='bottom', fontsize=8)

ax.set_xticks(range(n))
ax.set_xticklabels(page_labels_sorted_t, fontsize=7, rotation=45, ha='right'
ax.set_ylabel('PageRank Score (π)')
ax.set_title('PageRank Scores — G̃ with Teleportation (α = 0.85)\nPages ranke
ax.set_ylim(0, max(pi_tilde_sorted) * 1.15)
ax.grid(axis='y', alpha=0.3)

plt.tight_layout()
plt.savefig('pagerank_G_tilde.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: pagerank_G_tilde.png")
```




```
Saved: pagerank_G_tilde.png
```

Section 10 — Compare Rankings: G_fixed vs G̃

(Project Instruction Step 11)

We now compare the two PageRank vectors side by side.
Adding teleportation slightly redistributes importance: heavily-linked pages lose a small amount of weight (because 15% of visits are now uniform random), while isolated pages gain a tiny bit.
If the top pages remain the same, it confirms the link structure is the dominant factor in determining importance.

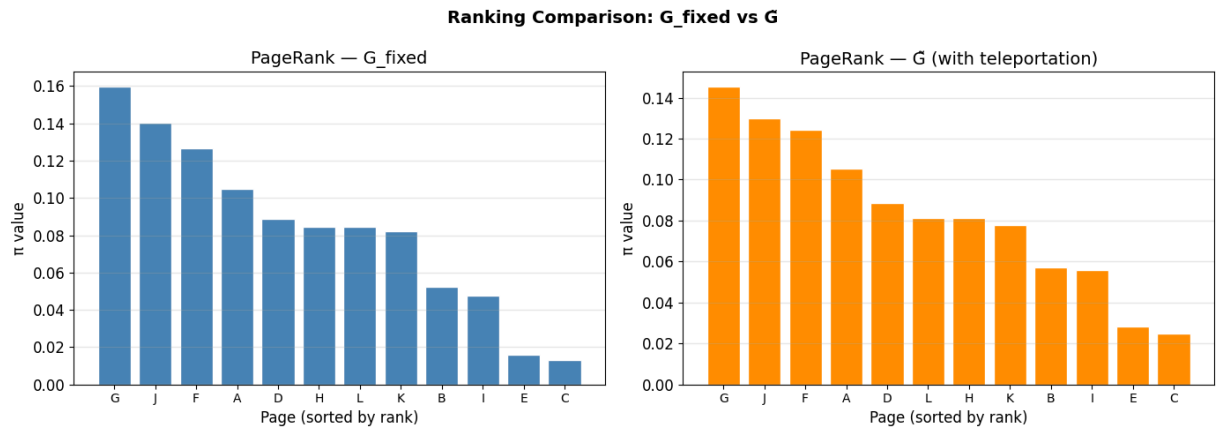
In [17]:
```
# Side-by-side ranking comparison table
rank_fixed_arr = np.argsort(-pi_fixed)
```

```
rank_tilde_arr = np.argsort(-pi_tilde)

print("RANKING COMPARISON: G_fixed vs G_tilde")
print(f"{'Rank':<6} {'G_fixed page':<14} {'π (G_fixed)':<14} {'G_tilde page'
print("-" * 72)
for r in range(n):
    i = rank_fixed_arr[r]
    j = rank_tilde_arr[r]
    same = "✓" if i == j else "←changed"
    print(f"  #{r+1:<4} {labels[i]:<14} {pi_fixed[i]:<14.6f} {labels[j]:<14}
```

```
RANKING COMPARISON: G_fixed vs G_tilde
Rank   G_fixed page   π (G_fixed)    G_tilde page   π (G_tilde)    Same?
------------------------------------------------------------------------
  #1    G              0.159855       G              0.145502       ✓
  #2    J              0.140175       J              0.129794       ✓
  #3    F              0.126678       F              0.124250       ✓
  #4    A              0.104962       A              0.105515       ✓
  #5    D              0.088674       D              0.088495       ✓
  #6    H              0.084301       L              0.081375       ←changed
  #7    L              0.084301       H              0.081030       ←changed
  #8    K              0.082114       K              0.078004       ✓
  #9    B              0.052481       B              0.057344       ✓
  #10   I              0.047504       I              0.055719       ✓
  #11   E              0.015835       E              0.028287       ✓
  #12   C              0.013120       C              0.024686       ✓
```

In [18]:

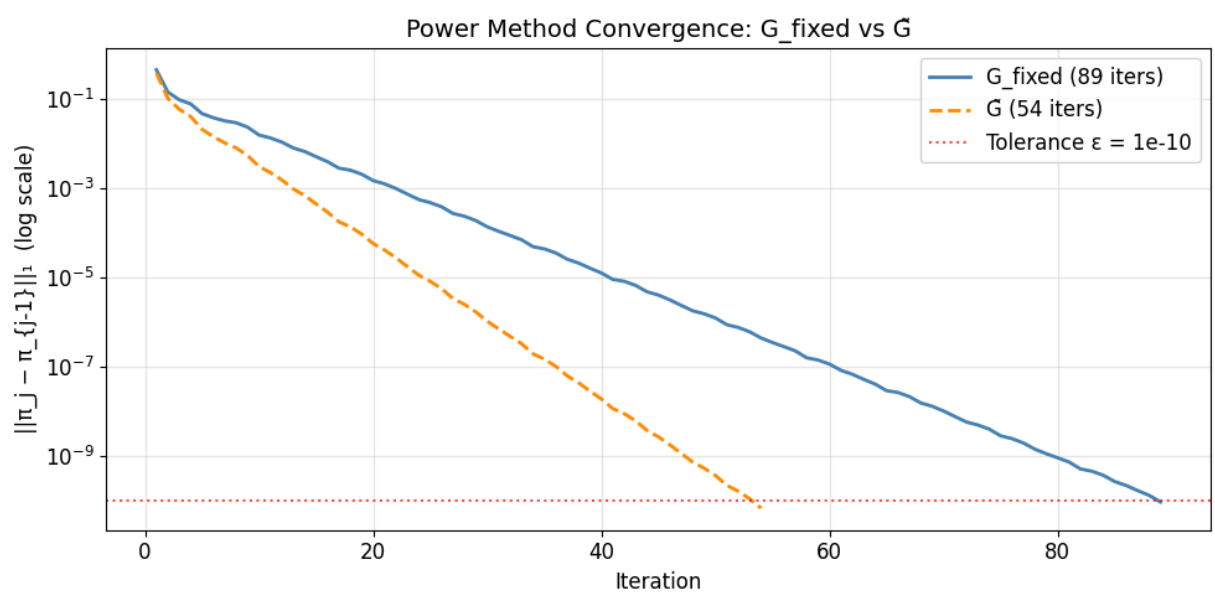
```
# Side-by-side bar chart comparison
# Slide 17

fig, axes = plt.subplots(1, 2, figsize=(14, 5), sharey=False)

for ax, pi_vals, color, title in [
    (axes[0], pi_fixed, 'steelblue', 'G_fixed'),
    (axes[1], pi_tilde, 'darkorange', 'G̃ (with teleportation)'),
]:
    idx = np.argsort(-pi_vals)
    ax.bar(range(n), pi_vals[idx], color=color, edgecolor='white')
    ax.set_xticks(range(n))
    ax.set_xticklabels([labels[i] for i in idx], fontsize=10)
    ax.set_title(f'PageRank — {title}')
    ax.set_ylabel('π value')
    ax.set_xlabel('Page (sorted by rank)')
    ax.grid(axis='y', alpha=0.3)

plt.suptitle('Ranking Comparison: G_fixed vs G̃', fontsize=14, fontweight='bo
plt.tight_layout()
plt.savefig('ranking_comparison.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: ranking_comparison.png")
```




```
Saved: ranking_comparison.png
```

In [19]:

```python
# Combined convergence plot (G_fixed and G_tilde together)
# Slide 15.

fig, ax = plt.subplots(figsize=(10, 5))

ax.semilogy(range(1, len(hist_fixed) + 1), hist_fixed,
            color='steelblue', linewidth=2, label=f'G_fixed ({iters_fixed} i
ax.semilogy(range(1, len(hist_tilde) + 1), hist_tilde,
            color='darkorange', linewidth=2, linestyle='--',
            label=f'G̃ ({iters_tilde} iters)')

ax.axhline(y=1e-10, color='red', linestyle=':', alpha=0.7, label='Tolerance
ax.set_xlabel('Iteration')
ax.set_ylabel('||π_j - π_{j-1}||₁  (log scale)')
ax.set_title('Power Method Convergence: G_fixed vs G̃')
ax.legend()
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('convergence_both.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: convergence_both.png")
```



```
Saved: convergence_both.png
```

Section 11 — Term-Document Matrix T

(Project Instruction Step 12)

What T is:
T is a **17 × 12** binary matrix (17 keywords × 12 pages).

$$T_{kp} = \begin{cases} 1 & \text{if keyword } k \text{ appears on page } p \\ 0 & \text{otherwise} \end{cases}$$

How it's used (Saito Lec 25, Slide 11):
For a query vector q (binary, length 17), compute:

$$d = T^T q$$

d is a 12-element vector where d_p counts how many query keywords appear on page p.

Source: Fill in from the `TermDoc03` keyword PDF. Every row below corresponds to one keyword; every column to one page (A=0 ... L=11).

In [20]:
```
# Keywords and T matrix
# Source: TermDoc03 PDF — "Final Project WebPage KeyWords #3: Trees"


keywords = [
    'Ash',       # 0
    'Aspen',     # 1
    'Butternut', # 2
    'Cherry',    # 3
    'Elm',       # 4
    'Fir',       # 5
    'Ginkgo',    # 6
    'Hickory',   # 7
    'Katsura',   # 8
    'Magnolia',  # 9
    'Oak',       # 10
    'Pine',      # 11
    'Redwood',   # 12
    'Sassafras', # 13
    'Spruce',    # 14
    'Teak',      # 15
    'Willow',    # 16
]

# Quick lookup: keyword name → row index in T
kw = {k: i for i, k in enumerate(keywords)}

# T[keyword_index][page_index] = 1 if keyword on page, else 0
```

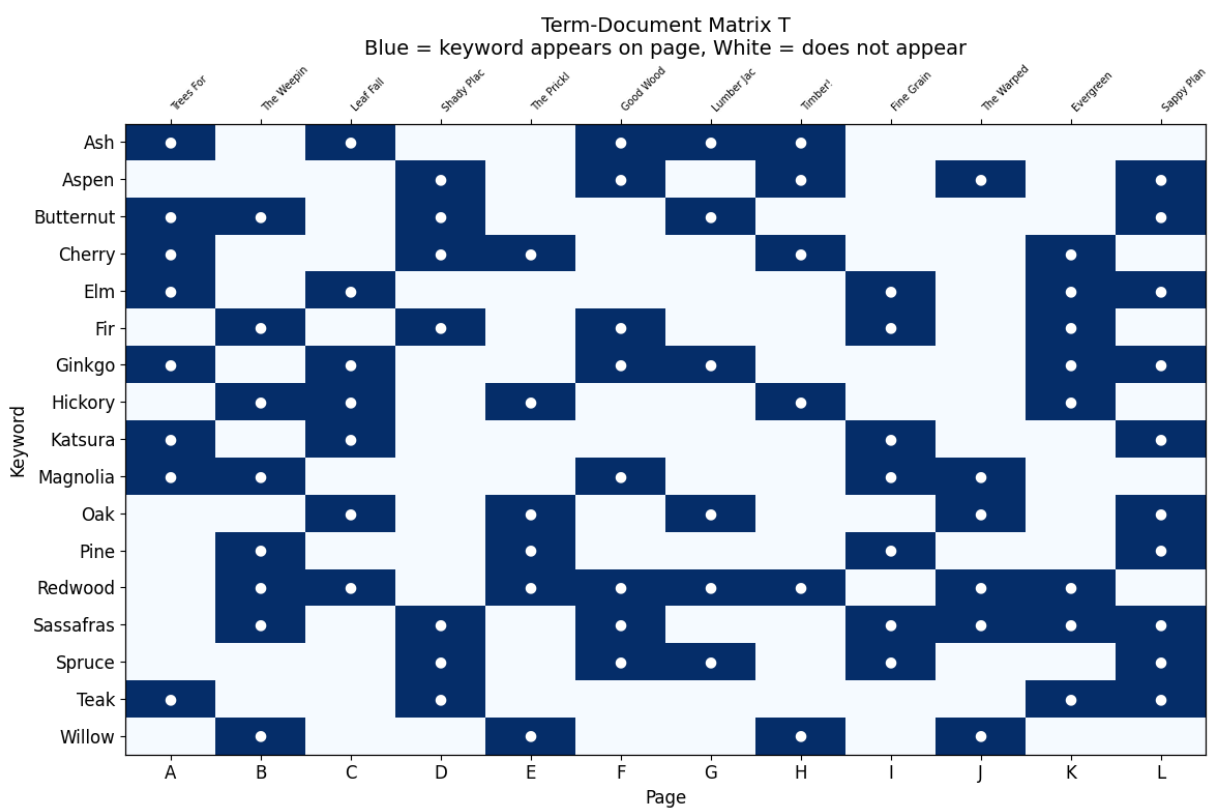
```
# Pages:     A  B  C  D  E  F  G  H  I  J  K  L
# Indices:   0  1  2  3  4  5  6  7  8  9 10 11
T = np.array([
  # A  B  C  D  E  F  G  H  I  J  K  L
  [ 1, 0, 1, 0, 0, 1, 1, 1, 0, 0, 0, 0 ],  # Ash    (A, C, F, G, H)
  [ 0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1 ],  # Aspen  (D, F, H, J, L)
  [ 1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1 ],  # Butternut (A, B, D, G, L)
  [ 1, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0 ],  # Cherry (A, D, E, H, K)
  [ 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 1 ],  # Elm    (A, C, I, K, L)
  [ 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 0 ],  # Fir    (B, D, F, I, K)
  [ 1, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 1 ],  # Ginkgo (A, C, F, G, K, L)
  [ 0, 1, 1, 0, 1, 0, 0, 1, 0, 0, 1, 0 ],  # Hickory (B, C, E, H, K)
  [ 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1 ],  # Katsura (A, C, I, L)
  [ 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0 ],  # Magnolia (A, B, F, I, J)
  [ 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1 ],  # Oak    (C, E, G, J, L)
  [ 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1 ],  # Pine   (B, E, I, L)
  [ 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0 ],  # Redwood (B, C, E, F, G, H, J, K
  [ 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 1, 1 ],  # Sassafras (B, D, F, I, J, K, L)
  [ 0, 0, 0, 1, 0, 1, 1, 0, 1, 0, 0, 1 ],  # Spruce (D, F, G, I, L)
  [ 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1 ],  # Teak   (A, D, K, L)
  [ 0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0 ],  # Willow (B, E, H, J)
], dtype=float)

print(f"T matrix shape: {T.shape}  (expected 17 rows × 12 columns)")
print(f"Number of keywords: {len(keywords)}")
print(f"Number of pages:    {n}")
print(f"\nKeywords per page:")
for p in range(n):
    kws_on_page = [keywords[k] for k in range(len(keywords)) if T[k, p] == 1
    print(f"  {labels[p]} ({page_titles[labels[p]]}): {', '.join(kws_on_page
```

```
T matrix shape: (17, 12)  (expected 17 rows × 12 columns)
Number of keywords: 17
Number of pages:    12

Keywords per page:
  A (Trees For All): Ash, Butternut, Cherry, Elm, Ginkgo, Katsura, Magnolia,
Teak
  B (The Weeping Willow): Butternut, Fir, Hickory, Magnolia, Pine, Redwood,
Sassafras, Willow
  C (Leaf Fall): Ash, Elm, Ginkgo, Hickory, Katsura, Oak, Redwood
  D (Shady Place): Aspen, Butternut, Cherry, Fir, Sassafras, Spruce, Teak
  E (The Prickly Pine): Cherry, Hickory, Oak, Pine, Redwood, Willow
  F (Good Wood): Ash, Aspen, Fir, Ginkgo, Magnolia, Redwood, Sassafras, Spru
ce
  G (Lumber Jack Man): Ash, Butternut, Ginkgo, Oak, Redwood, Spruce
  H (Timber!): Ash, Aspen, Cherry, Hickory, Redwood, Willow
  I (Fine Grains): Elm, Fir, Katsura, Magnolia, Pine, Sassafras, Spruce
  J (The Warped Board): Aspen, Magnolia, Oak, Redwood, Sassafras, Willow
  K (Evergreen Plants): Cherry, Elm, Fir, Ginkgo, Hickory, Redwood, Sassafra
s, Teak
  L (Sappy Planks): Aspen, Butternut, Elm, Ginkgo, Katsura, Oak, Pine, Sassa
fras, Spruce, Teak
```

In [21]:

```
# Heatmap of T matrix
# Slide 18
```

```
fig, ax = plt.subplots(figsize=(12, 8))

im = ax.imshow(T, cmap='Blues', aspect='auto', vmin=0, vmax=1)

ax.set_xticks(range(n))
ax.set_xticklabels([f"{labels[p]}" for p in range(n)])
ax.set_yticks(range(len(keywords)))
ax.set_yticklabels(keywords)
ax.set_xlabel('Page')
ax.set_ylabel('Keyword')
ax.set_title('Term-Document Matrix T\nBlue = keyword appears on page, White

# Mark each 1 with a dot for clarity
for k in range(len(keywords)):
    for p in range(n):
        if T[k, p] == 1:
            ax.text(p, k, '●', ha='center', va='center', color='white', font

# Add page title labels at top
ax2 = ax.twiny()
ax2.set_xlim(ax.get_xlim())
ax2.set_xticks(range(n))
ax2.set_xticklabels([page_titles[labels[p]][:10] for p in range(n)],
                    rotation=45, ha='left', fontsize=7)

plt.tight_layout()
plt.savefig('term_document_matrix.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: term_document_matrix.png")
```




```
Saved: term_document_matrix.png
```

Section 12 — Search Queries

(Project Instruction Steps 13–15)

The process for every query:

1. Build query vector q (length 17, one entry per keyword)
2. Compute $d = T^T q$ — a 12-element responsiveness vector
3. Apply Boolean logic to determine the **responsive set**
4. Sort responsive pages by $\tilde{G}$ PageRank (descending)

The four query types:

Query	Logic	Responsive if...
SINGLE	Page contains the keyword	`d[p] > 0`
AND	Page contains **all** query terms	`d[p] == number of terms`
OR	Page contains **at least one** term	`d[p] >= 1`
BUT NOT	Page has term A and **not** term B	`T[A][p]==1 and T[B][p]==0`

In [22]:
```
def show_query_results(query_name, responsive_indices, pi_tilde, labels, pag
    """
    Print the final ranked results for a query.
    Responsive pages are sorted by G_tilde PageRank (descending).
    """
    # Sort responsive pages by PageRank (descending)
    ranked = sorted(responsive_indices, key=lambda i: -pi_tilde[i])

    print(f"\n{'='*60}")
    print(f"  QUERY: {query_name}")
    print(f"{'='*60}")

    if not ranked:
        print("  → No pages match this query.")
        return ranked

    print(f"  {'Result':<8} {'Page':<6} {'Title':<25} {'PageRank'}")
    print(f"  {'-'*58}")
    for result_rank, i in enumerate(ranked, start=1):
        print(f"  #{result_rank:<7} {labels[i]:<6} {page_titles[labels[i]]:<
    return ranked
```

In [23]:
```
# Query 1: SINGLE — "Elm"
#
# Query vector q has a 1 only at the index of 'Elm'.
# d = T^T @ q  →  d[p] = 1 if 'Elm' is on page p, else 0.
# Responsive set: pages where d[p] > 0.
```

```
q1 = np.zeros(len(keywords))
q1[kw['Elm']] = 1

d1 = T.T @ q1   # shape (12,)  —  d1[p] = # query keywords on page p

print("Query vector q1 (non-zero entries):")
for k, v in enumerate(q1):
    if v > 0:
        print(f"  q1[{keywords[k]}] = {v}")

print(f"\nd = T^T @ q1  (responsiveness scores):")
for p in range(n):
    print(f"  Page {labels[p]}: d = {d1[p]:.0f}  {'← responsive' if d1[p] >

responsive_1 = [p for p in range(n) if d1[p] > 0]
results_1 = show_query_results('SINGLE: Elm', responsive_1, pi_tilde, labels
```

```
Query vector q1 (non-zero entries):
  q1[Elm] = 1.0

d = T^T @ q1  (responsiveness scores):
  Page A: d = 1  ← responsive
  Page B: d = 0
  Page C: d = 1  ← responsive
  Page D: d = 0
  Page E: d = 0
  Page F: d = 0
  Page G: d = 0
  Page H: d = 0
  Page I: d = 1  ← responsive
  Page J: d = 0
  Page K: d = 1  ← responsive
  Page L: d = 1  ← responsive

============================================================
  QUERY: SINGLE: Elm
============================================================
  Result   Page   Title                     PageRank
  ------------------------------------------------------------
  #1       A      Trees For All             0.105515
  #2       L      Sappy Planks              0.081375
  #3       K      Evergreen Plants          0.078004
  #4       I      Fine Grains               0.055719
  #5       C      Leaf Fall                 0.024686
```

In [24]:

```
# Query 2: AND — "Aspen AND Sassafras"
#
# Both terms set to 1 in query vector.
# d[p] = 2 means the page has BOTH Aspen AND Sassafras.
# d[p] = 1 means only one of the two — NOT responsive for AND.
# d[p] = 0 means neither — not responsive.

q2 = np.zeros(len(keywords))
q2[kw['Aspen']] = 1
q2[kw['Sassafras']] = 1
```

```
d2 = T.T @ q2
num_terms_2 = int(q2.sum())   # = 2; ALL terms must appear for AND

print("Query vector q2 (non-zero entries):")
for k, v in enumerate(q2):
    if v > 0:
        print(f"  q2[{keywords[k]}] = {v}")

print(f"\nd = T^T @ q2  (AND requires d[p] == {num_terms_2}):")
for p in range(n):
    status = '← BOTH terms present (responsive!)' if d2[p] == num_terms_2 el
             f'← only {int(d2[p])}/2 terms' if d2[p] > 0 else ''
    print(f"  Page {labels[p]}: d = {d2[p]:.0f}  {status}")

responsive_2 = [p for p in range(n) if d2[p] == num_terms_2]
results_2 = show_query_results('AND: Aspen AND Sassafras', responsive_2, pi_
```

```
Query vector q2 (non-zero entries):
  q2[Aspen] = 1.0
  q2[Sassafras] = 1.0

d = T^T @ q2  (AND requires d[p] == 2):
  Page A: d = 0
  Page B: d = 1  ← only 1/2 terms
  Page C: d = 0
  Page D: d = 2  ← BOTH terms present (responsive!)
  Page E: d = 0
  Page F: d = 2  ← BOTH terms present (responsive!)
  Page G: d = 0
  Page H: d = 1  ← only 1/2 terms
  Page I: d = 1  ← only 1/2 terms
  Page J: d = 2  ← BOTH terms present (responsive!)
  Page K: d = 1  ← only 1/2 terms
  Page L: d = 2  ← BOTH terms present (responsive!)

============================================================
  QUERY: AND: Aspen AND Sassafras
============================================================
  Result   Page   Title                      PageRank
  ------------------------------------------------------------
  #1       J      The Warped Board           0.129794
  #2       F      Good Wood                  0.124250
  #3       D      Shady Place                0.088495
  #4       L      Sappy Planks               0.081375
```

In [25]:

```
# Query 3: OR — "Willow OR Redwood"
#
# Both terms set to 1 in query vector.
# d[p] >= 1 means the page has at least one of Willow, Redwood.
# d[p] = 2 means it has BOTH (still responsive — more is fine for OR).

q3 = np.zeros(len(keywords))
q3[kw['Willow']] = 1
q3[kw['Redwood']] = 1
```

```
d3 = T.T @ q3

print("Query vector q3 (non-zero entries):")
for k, v in enumerate(q3):
    if v > 0:
        print(f"  q3[{keywords[k]}] = {v}")

print(f"\nd = T^T @ q3  (OR requires d[p] >= 1):")
for p in range(n):
    kws_present = []
    if T[kw['Willow'], p] == 1: kws_present.append('Willow')
    if T[kw['Redwood'], p] == 1: kws_present.append('Redwood')
    status = f'← has: {kws_present}' if kws_present else ''
    print(f"  Page {labels[p]}: d = {d3[p]:.0f}  {status}")

responsive_3 = [p for p in range(n) if d3[p] >= 1]
results_3 = show_query_results('OR: Willow OR Redwood', responsive_3, pi_til
```

```
Query vector q3 (non-zero entries):
  q3[Redwood] = 1.0
  q3[Willow] = 1.0

d = T^T @ q3  (OR requires d[p] >= 1):
  Page A: d = 0
  Page B: d = 2  ← has: ['Willow', 'Redwood']
  Page C: d = 1  ← has: ['Redwood']
  Page D: d = 0
  Page E: d = 2  ← has: ['Willow', 'Redwood']
  Page F: d = 1  ← has: ['Redwood']
  Page G: d = 1  ← has: ['Redwood']
  Page H: d = 2  ← has: ['Willow', 'Redwood']
  Page I: d = 0
  Page J: d = 2  ← has: ['Willow', 'Redwood']
  Page K: d = 1  ← has: ['Redwood']
  Page L: d = 0

============================================================
  QUERY: OR: Willow OR Redwood
============================================================
  Result   Page   Title                      PageRank
  ------------------------------------------------------------
  #1       G      Lumber Jack Man           0.145502
  #2       J      The Warped Board          0.129794
  #3       F      Good Wood                 0.124250
  #4       H      Timber!                   0.081030
  #5       K      Evergreen Plants          0.078004
  #6       B      The Weeping Willow        0.057344
  #7       E      The Prickly Pine          0.028287
  #8       C      Leaf Fall                 0.024686
```

In [26]:

```
# Query 4: BUT NOT — "Elm BUT NOT Fir"

# Must have Elm. Must NOT have Fir.
# We handle this with explicit set operations rather than a single query vec
# because a negative query term cannot be encoded directly in d = T^T q.
# (Subtracting Fir in q would give d[p] = -1 for Fir-only pages, not useful.
```

```
has_elm = set(p for p in range(n) if T[kw['Elm'], p] == 1)
has_fir = set(p for p in range(n) if T[kw['Fir'], p] == 1)

print(f"Pages with 'Elm':     {sorted([labels[p] for p in has_elm])}")
print(f"Pages with 'Fir':     {sorted([labels[p] for p in has_fir])}")
print(f"Pages with Elm and NOT Fir: {sorted([labels[p] for p in has_elm - ha

responsive_4 = sorted(has_elm - has_fir)   # Python set difference
results_4 = show_query_results('BUT NOT: Elm BUT NOT Fir', responsive_4, pi_
```

```
Pages with 'Elm':     ['A', 'C', 'I', 'K', 'L']
Pages with 'Fir':     ['B', 'D', 'F', 'I', 'K']
Pages with Elm and NOT Fir: ['A', 'C', 'L']

============================================================
  QUERY: BUT NOT: Elm BUT NOT Fir
============================================================
  Result   Page   Title                     PageRank
  ------------------------------------------------------------
  #1       A      Trees For All             0.105515
  #2       L      Sappy Planks              0.081375
  #3       C      Leaf Fall                 0.024686
```

Section 13 — Complete Summary of All Results

This final section prints a clean summary of every result from the project — ready to read off for your presentation.

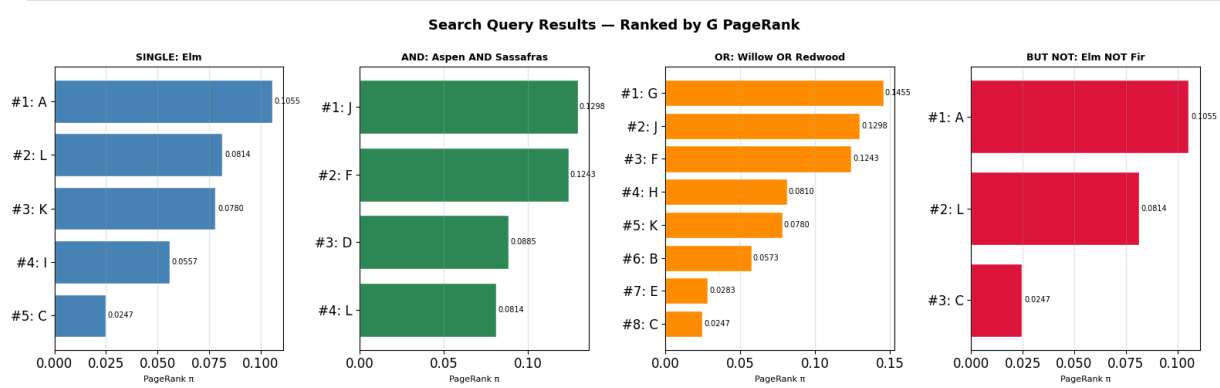
In [27]:
```
print("=" * 70)
print("  MAT 167 FINAL PROJECT — COMPLETE RESULTS SUMMARY")
print("  Google Network #10  |  Keywords: Trees Set #3")
print("=" * 70)

# PageRank Rankings
print("\n—— PageRank Rankings ——")
print(f"{'Rank':<6} {'G_fixed page':<16} {'π(G_fixed)':<14} {'G̃ page':<16} {
print("-" * 65)
for r in range(n):
    i = rank_fixed_arr[r]
    j = sorted_tilde[r]
    print(f"  #{r+1:<4} {labels[i]+': '+page_titles[labels[i]]:<16} {pi_fixe
          f" {labels[j]+': '+page_titles[labels[j]]:<16} {pi_tilde[j]:.6f}")

# Power Method
print(f"\n—— Power Method ——")
print(f"  G_fixed:  converged in {iters_fixed} iterations")
print(f"  G̃:        converged in {iters_tilde} iterations")
print(f"  Results match eigendecomposition: ✓")

# Search Queries
print("\n—— Search Query Results (ranked by G̃ PageRank) ——")
```

```
for name, results in [
    ('SINGLE: Elm',                 results_1),
    ('AND: Aspen AND Sassafras',     results_2),
    ('OR: Willow OR Redwood',        results_3),
    ('BUT NOT: Elm BUT NOT Fir',     results_4),
]:
    print(f"\n  {name}:")
    if not results:
        print("    → No matching pages.")
    else:
        for rank, i in enumerate(results, 1):
            print(f"    #{rank}: {labels[i]} — {page_titles[labels[i]]}  (π

print("\n" + "=" * 70)
print("  All computations complete.")
print("=" * 70)
```

```
======================================================================
  MAT 167 FINAL PROJECT — COMPLETE RESULTS SUMMARY
  Google Network #10  |  Keywords: Trees Set #3
======================================================================

─── PageRank Rankings ───
Rank   G_fixed page     π(G_fixed)     G̃ page          π(G̃)
-------------------------------------------------------------------
  #1    G: Lumber Jack Man 0.159855       G: Lumber Jack Man 0.145502
  #2    J: The Warped Board 0.140175       J: The Warped Board 0.129794
  #3    F: Good Wood     0.126678       F: Good Wood     0.124250
  #4    A: Trees For All 0.104962       A: Trees For All 0.105515
  #5    D: Shady Place   0.088674       D: Shady Place   0.088495
  #6    H: Timber!       0.084301       L: Sappy Planks  0.081375
  #7    L: Sappy Planks  0.084301       H: Timber!       0.081030
  #8    K: Evergreen Plants 0.082114       K: Evergreen Plants 0.078004
  #9    B: The Weeping Willow 0.052481       B: The Weeping Willow 0.057344
  #10   I: Fine Grains   0.047504       I: Fine Grains   0.055719
  #11   E: The Prickly Pine 0.015835       E: The Prickly Pine 0.028287
  #12   C: Leaf Fall     0.013120       C: Leaf Fall     0.024686

─── Power Method ───
  G_fixed:  converged in 89 iterations
  G̃:        converged in 54 iterations
  Results match eigendecomposition: ✓

─── Search Query Results (ranked by G̃ PageRank) ───

  SINGLE: Elm:
    #1: A — Trees For All  (π = 0.105515)
    #2: L — Sappy Planks  (π = 0.081375)
    #3: K — Evergreen Plants  (π = 0.078004)
    #4: I — Fine Grains  (π = 0.055719)
    #5: C — Leaf Fall  (π = 0.024686)

  AND: Aspen AND Sassafras:
    #1: J — The Warped Board  (π = 0.129794)
    #2: F — Good Wood  (π = 0.124250)
    #3: D — Shady Place  (π = 0.088495)
    #4: L — Sappy Planks  (π = 0.081375)

  OR: Willow OR Redwood:
    #1: G — Lumber Jack Man  (π = 0.145502)
    #2: J — The Warped Board  (π = 0.129794)
    #3: F — Good Wood  (π = 0.124250)
    #4: H — Timber!  (π = 0.081030)
    #5: K — Evergreen Plants  (π = 0.078004)
    #6: B — The Weeping Willow  (π = 0.057344)
    #7: E — The Prickly Pine  (π = 0.028287)
    #8: C — Leaf Fall  (π = 0.024686)

  BUT NOT: Elm BUT NOT Fir:
    #1: A — Trees For All  (π = 0.105515)
    #2: L — Sappy Planks  (π = 0.081375)
    #3: C — Leaf Fall  (π = 0.024686)
```

```
======================================================================
  All computations complete.
======================================================================
```

In [28]:
```
# Final combined figure: all 4 query results as a visual summary


queries = [
    ('SINGLE: Elm',              results_1),
    ('AND: Aspen AND Sassafras', results_2),
    ('OR: Willow OR Redwood',    results_3),
    ('BUT NOT: Elm NOT Fir',     results_4),
]

fig, axes = plt.subplots(1, 4, figsize=(16, 5))
colors = ['steelblue', 'seagreen', 'darkorange', 'crimson']

for ax, (name, results), color in zip(axes, queries, colors):
    if not results:
        ax.text(0.5, 0.5, 'No results', ha='center', va='center', transform=
        ax.set_title(name, fontsize=9, fontweight='bold')
        continue

    page_names = [labels[i] for i in results]
    page_scores = [pi_tilde[i] for i in results]

    bars = ax.barh(range(len(results)), page_scores, color=color, edgecolor=
    ax.set_yticks(range(len(results)))
    ax.set_yticklabels([f"#{r+1}: {labels[i]}" for r, i in enumerate(results
    ax.invert_yaxis()    # rank 1 at top
    ax.set_xlabel('PageRank π', fontsize=8)
    ax.set_title(name, fontsize=9, fontweight='bold')
    ax.grid(axis='x', alpha=0.3)

    for bar, val in zip(bars, page_scores):
        ax.text(val + 0.001, bar.get_y() + bar.get_height()/2,
                f'{val:.4f}', va='center', fontsize=7)

plt.suptitle('Search Query Results — Ranked by G̃ PageRank', fontsize=13, for
plt.tight_layout()
plt.savefig('all_query_results.png', dpi=150, bbox_inches='tight')
plt.show()
print("Saved: all_query_results.png")
```




```
Saved: all_query_results.png
```

Files Generated by This Notebook

The following `.png` files are saved in the same directory as this notebook.
Use them directly in your PowerPoint/Keynote slides:

File	Use for slide
`G_raw_heatmap.png`	Slide 13 — the Google matrix
`pagerank_fixed_G.png`	Slide 14 — PageRank bar chart (G_fixed)
`convergence_G_fixed.png`	Slide 15 — Power method convergence
`pagerank_G_tilde.png`	Slide 16 — PageRank bar chart ($\tilde{G}$)
`ranking_comparison.png`	Slide 17 — G vs $\tilde{G}$ side by side
`convergence_both.png`	Slide 15 — G_fixed vs $\tilde{G}$ convergence
`term_document_matrix.png`	Slide 18 — T matrix heatmap
`all_query_results.png`	Slides 19–22 — all 4 query results

Key Equations Quick Reference

Equation	Meaning
$G_{ij} = 1/\lvert P_i \rvert$ if $i \to j$ else 0	Raw Google matrix definition
$\pi^T G = \pi^T$	PageRank = left eigenvector of G, eigenvalue 1
$G^T \pi = \pi$	Equivalent form used in `eig(G.T)`
$\pi_j^T = \pi_{j-1}^T G$, then normalize	Power iteration update rule
$\lvert \pi \rvert_1 = \sum_i \pi_i = 1$	L1 normalization condition
$\tilde{G} = 0.85G + 0.15E$	Teleportation-adjusted Google matrix
$E = \frac{1}{n}\mathbf{1}\mathbf{1}^T$	Uniform teleportation

Equation	Meaning
	matrix
$d = T^T q$	Query responsiveness vector